

Pixel-Level Collision-Free Grasp Prediction Network for Medical Test Tube Sorting on Cluttered Trays

Shihao Ge , Beiping Hou , Wen Zhu , Yuzhen Zhu , Senjian Lu , and Yangbin Zheng 

Abstract—Robotic sorting shows a promising aspect for future developments in medical field. However, vision-based grasp detection of medical devices is usually in unstructured or cluttered environments, which raises major challenges for the development of robotic sorting systems. In this letter, a pixel-level grasp detection method is proposed to predict the optimal collision-free grasp configuration on RGB images. First, an Adaptive Grasp Flex Classify (AGFC) model is introduced to add category attributes to distinguish test tube arrangements in complex scenarios. Then, we propose an end-to-end trainable CNN-based architecture, which delivers high quality results for grasp detection and avoids the confusion in neural network learning, to generate the AGFC-model. Utilizing this, we design a Residual Efficient Atrous Spatial Pyramid (REASP) block to further increase the accuracy of grasp detection. Finally, a collision-free manipulation policy is designed to guide the robot to grasp. Experiments on various scenarios are implemented to illustrate the robustness and the effectiveness of our approach, and a robotic grasping platform is constructed to evaluate its application performance. Overall, the developed robotic sorting system achieves a success rate of 95% on test tube sorting.

Index Terms—Collision-free grasp detection, deep learning in grasping and manipulation, medical robots and systems.

I. INTRODUCTION

AUTOMATING the medical device sorting process has the potential to significantly address the safety and efficiency issues in medical practice [1], [2]. In many clinical wards and emergency rooms in China, post-assay test tubes are often cluttered in trays. Replacing manual sorting with robots has several benefits, including increased efficiency, reduced risk of contamination and the ability to record and track data. However, extracting semantic information from visual inputs to find a grasp

configuration that satisfies the criteria of a specific grasping task, especially in cluttered environments, is still a challenging problem [3], [4], [5]. To sort the cluttered arrangement of medical test tubes on trays, several challenges need to be addressed. The dense and unordered arrangement of test tubes on the tray imposes strict requirements on grasp detection accuracy. In this case, an incorrect grasping order can potentially lead to unsafe grasping actions, such as collisions between the gripper and the environment or neighboring test tubes, which increases risk of damage to test tubes and grippers. Furthermore, the transparency of glass test tubes and the optical reflectance of plastic trays pose a challenge to many visual perception algorithms which base on depth or point cloud information.

The planar grasp representation introduced by Kumra et al. [6] formulates the problem of robotic grasp detection as finding an optimal grasp configuration given a depth image. However, this approach may not be suitable for grasp problems in cluttered scenes where objects are closely packed together. In such scenarios, the gripper is prone to collisions during the grasping process. Robotic grasping for the tube sorting application has to be fast and flexible in order to be used efficiently. With this goal in mind, our approach, similar to previous works [7], [8], [9], [10], [11], considers planar grasping by parameterizing the action space using 3D gripper position and planar orientation, rather than 6-DOF grasping [12]. The main drawback of unrestricted 6-DOF grasps lies in the complexity of the data labeling process and the increased run-time [13], [14].

Based on this, we propose the AGFC-model, which improves the learning efficiency of the neural network for the grasp quality score by incorporating an additional grasp category attribute. In this model, we separate the prediction of grasp quality scores from the prediction of grasp categories, which enhances the robustness of grasp detection in varying scenarios. The AGFC-model essentially represents a set of contiguous graspable points with corresponding grasp attributes, which avoids a complicated annotation process.

Furthermore, we design a pixel-level grasp classification network to directly generate the AGFC-model given camera data, based on the generative approach by Morrison et al. [15], which overcomes the limitations of discrete sampling for grasp candidates and reduces computational complexity [9], [16], [17], resulting in improved efficiency. The proposed network takes RGB images as input to overcome the limitations imposed by the optically-challenging nature of tube sorting scenes. To enhance the robustness and efficiency of the network in unstructured scenes, the Residual Efficient Atrous Spatial Pyramid (REASP)

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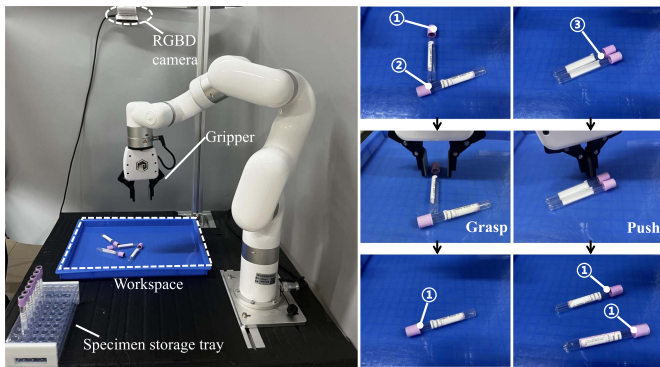


Fig. 1. Left: Setup for robot grasping experiments. In the workspace with N test tubes, the robot iteratively plans grasps from RGB images and attempts to lift and place the objects into the storage tray. Right: The three grasp categories include collision-free grasp points (1), collision-prone grasp points (2), and points that require a push action to grasp (3).

block is designed to extract multi-scale features and facilitate advanced analysis and interpretation of complex scenes. In addition, we add a grasp category head to assign richer semantic information to each point, which enhances the robot's capabilities in completing pushing and grasping tasks. Then, a collision-free grasping policy is proposed to process the AGFC-model and guide the real robot to perform grasping or pushing actions, as shown in Fig. 1. Finally, we contribute a test tube dataset containing about 1000 images and evaluate the performance of our system in different scenarios by performing grasping trials with a Xarm6 robot. The main contributions of this study are as follows:

- We study the sorting solutions of medical test tubes for robot grasping applications, and we propose an end-to-end trainable network architecture for grasp detection and pixel-level semantic segmentation, which makes grasp detection more efficient and accurate.
- The AGFC-model is proposed to represent different grasp categories and a collision-free manipulation policy is proposed to guide the robot to grasp, which is suitable for complex planar grasping tasks.
- We present a practical robotic grasping platform and create a medical test tube dataset with different scenes and 1000 samples. This dataset is used to perform experiments on grasp detection and real-world grasping, and to evaluate the performance of the robotic grasping application on this platform.

II. RELATED WORKS

A. Vision-Guided Robotic Grasping

Robotic grasping in cluttered environments has been a prominent area of research for several decades [18]. Sensors commonly used for perception in robotic environments include 3D vision systems or RGB-D cameras. Mahler et al. [9] proposed Dex-Net1.0 to 4.0, which predicts the probability of success for candidate grasps based on point cloud. Yan et al. [19] reconstructed full 3D point clouds as representation for robotic applications. Similar methods that use point clouds as input,

while achieving high prediction accuracy, often suffer from unsatisfactory prediction speed and poor performance when collecting point cloud data for optically-challenging objects. With the availability of affordable RGB-D sensors, a new line of research has emerged, which focuses on using visual features obtained from RGB or RGB-D images to predict grasp points [20], [21].

B. Grasp Representation

Grasp synthesis, an active field of research for many years, involves formulating a stable robotic grasp for a given object. Jiang et al. [22] proposed a 7-dimensional gripper configuration (i.e., the 3D location, 3D orientation and gripper opening width) to represent the final state of the gripper. Lenz et al. [23] introduced a 5-dimensional grasp representation, which reduces the computational cost by reducing the number of dimensions. Morrison et al. [15] moved away from the traditional rectangular representation and proposed a novel grasp representation that directly generates a grasp for each pixel, rather than sampling the input image to generate grasp candidates. These representations suffer from a lack of effective pixel points categorization, which may not be applicable to scenes with closely spaced objects. In this work, we propose a generative categorical grasp representation model to represent the grasping point information of test tubes in different arrangements.

C. Network

Recent approaches to robotic grasping do not rely on sampling of grasp candidates [9], [23], [24], but rather directly generate pixel-level grasp configuration, analogous to perform pixelwise semantic segmentation task [6], [15], [25]. When moving away from techniques that use discrete sampling of grasp candidates and towards these generative approaches, the computation times increase. Some works employ encoder/decoder-based CNN architectures or dilated convolutions [26] for the task of semantic segmentation. For instance, Araki et al. [27] proposed a network to achieve simultaneous execution of object detection and segmentation for a suction gripper. In this work, we propose an end-to-end CNN-based network that simultaneously predicts pushing points and grasping points, which is able to grasp without collision and place the test tubes into the tray.

III. METHOD

The main components of our approach consist of three parts: feature extractor, grasp detector and collision-free manipulation policy, as shown in Fig. 2. The feature extractor takes RGB images as input and fuses features at different scales. The grasp detector, where the extracted features are pixelated and predicted into grasp quality map, grasp category maps, grasp angle maps, and grasp width maps, are based on which the AGFC-model is further generated. Finally, the collision-free manipulation policy utilizes the AGFC-model, which contains rich semantic information, to guide the robot in executing optimal collision-free grasping or pushing actions.

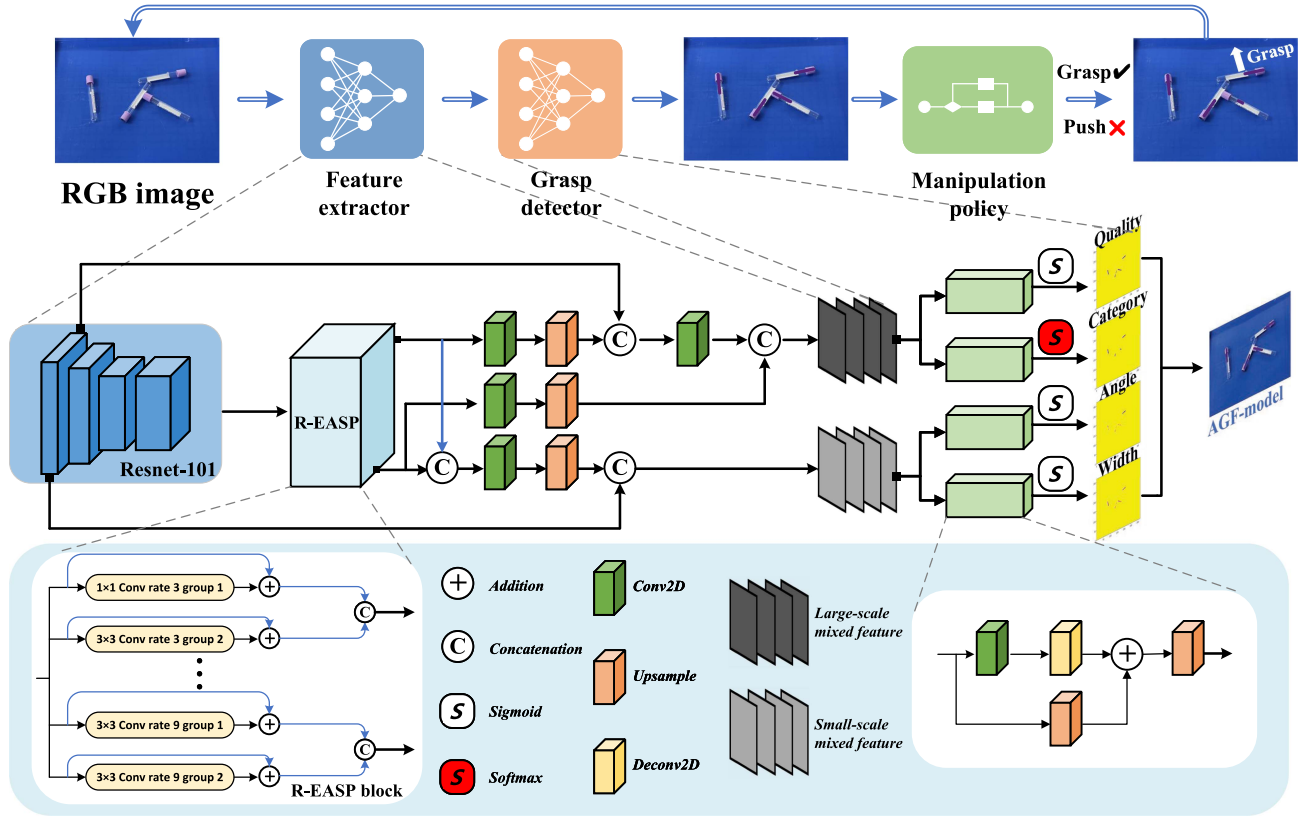


Fig. 2. Architecture of our approach.

A. Feature Extractor

In this letter, a feature extraction component is constructed based on the Deep-LabV3+ [28] to achieve accurate segmentation of tubes, which converts grasp detection problem in unstructured environments into a pixel-level segmentation problem based on RGB images. We use Resnet-101 [29] as the backbone to extract features from RGB images, and the extracted features are submitted to the Residual Efficient Atrous Spatial Pyramid (REASP) block.

1) *REASP-Block*: Chen et al. [26] proposed ASPP to capture objects and significant image context information at multiple scales. However, With the increasing of dilation rate, the receptive field increases, making the pixel loss in the computation on the atrous convolution. In contrast, we propose the REASP block with the 8-parallel structure to improve the utilization efficiency of information, where each pyramid is constructed using atrous convolutions with varying dilation rates ($rate = \{1, 6, 9, 12\}$) and group numbers ($num = \{1, 2\}$). This configuration enables the extraction of features at various scales, allowing for a more comprehensive capture of rich semantic information. We also make some improvements by using group convolution and skip connections. This reduces the number of parameters and effectively avoids gradient vanishing, leading to improved model accuracy, as demonstrated in the ablation studies in Section V-A.

2) *Fusion Module*: We employ the feature fusion module proposed by Wang et al. [25], which merges features from multiple levels in various orders. The feature fusion module

combines features from multiple levels, allowing for a more comprehensive representation of information.

B. Grasp Detector

Four task-specific sub-networks responsible for predicting grasp quality, category, angle and width are designed based on the extracted multi-scale fused features and connected in parallel to the decoder. We introduce residual connections at the output heads to retain more information. By inserting the upsampling and deconvolution module, the spatial information is gradually restored to capture the more refined target boundary of objects to achieve pixel-level prediction.

Quality head outputs the grasp quality score corresponding to each point, with fractional values ranging from 0 to 1. Category head outputs the grasp category ($C = \{1, 2, 3\}$) for each pixel, which classifies different test tube arrangements and facilitates the implementation of subsequent grasping policy. Angle head outputs k for each point (The computed grasp angle is given by $\theta = \frac{2\pi k}{120}$). Width head outputs the grasp width corresponding to each point.

C. Grasp Representation

Given a set of RGB observations, we aim to learn a grasp representation that is dense, semantically interpretable, and directly applicable to test tube manipulation.

1) *Problem Formulation*: In this subsection, we improve upon the traditional pixel-level grasp representation for handling

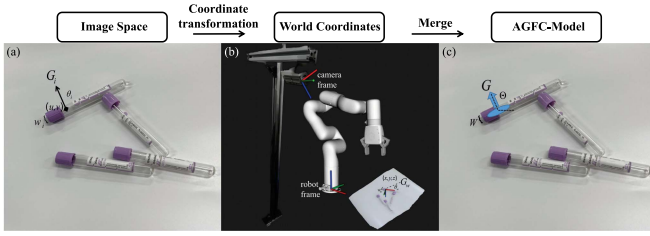


Fig. 3. Formation process of grasp representation model. The process involves mapping from the image coordinate system to the real-world coordinate system and ultimately generating pixel-level predictions.

complex test tube arrangements. For each pixel in a given RGB image $I = \mathbb{R}^{3 \times h \times w}$ with height h and width w , we assign an additional grasp category attribute to a grasp representation, which is described by

$$G_i = (u, v, \theta_i, w_i, q, c) \quad (1)$$

where (u, v) corresponds to the pixel coordinates of the grasp point, θ_i and w_i represent the grasp angle and the grasp width in the camera's reference frame respectively, q is the grasp confidence score, and c is the grasp category (Fig. 3(a)).

Similarly, we define the gripper configuration in the world coordinate system as

$$G_w = (p, \theta_w, w_w, q, c) \quad (2)$$

where $p(x, y, z)$ is the gripper's center position, θ_w is the gripper's rotation angle along the z-axis, w_w is the required width for the gripper, q and c are the same as in (1).

In order to execute the grasp point obtained in image space on the robot, we apply the following transformation to convert the image coordinates into the world coordinates (Fig. 3(b)).

$$G_w = \mathcal{T}_{wc}(\mathcal{T}_{ci}(G_i)) \quad (3)$$

where $\mathcal{T}_{wc}$ transforms from the camera frame to the world frame and $\mathcal{T}_{ci}$ transforms from 2D image coordinates to the 3D camera frame.

2) *Adaptive Grasp Flex Classify Model*: Determining the grasp positions of the test tubes on the cluttered tray is a key challenge. Obstacles include potential collisions between adjacent test tubes and the gripper due to incorrect grasping order. In addition, the parallel arrangement of test tubes poses a challenge by making them impossible to grasp. To address these challenges, we add a one-dimensional vector to the original grasp representation to classify each grasp candidate, which matches our added output header. We merge all gripper configurations in the robot coordinate system to obtain the AGFC-model, as shown in Fig. 3(c). This can be formulated as:

$$G = (\Theta, W, Q, C) \quad (4)$$

where Θ , W , Q and C represent the set of grasp angle, grasp width, grasp quality score and grasp category respectively calculated at each pixel of an image using the (2).

Algorithm 1: Collision-Free Manipulation Policy.

input : $G = (\Theta, W, Q, C) \leftarrow$ the AGFC-model

while $Flag \leftarrow True$ **do**

 Update input

$P_{all} \leftarrow F_{thr}(Q, a)$

$P_1, P_2, P_3 = \text{getIndex}(\text{argmax} C)$

$P_{grasp} \leftarrow P_{all} \cap P_1$

$P_{push} \leftarrow P_{all} \cap P_3$

if $P_{grasp} \neq \phi$ **then**

$P_{grasp} \mapsto G_{grasp}$

$G_{grasp}^* = \text{argmax}_{G_{grasp}} Q$

$\text{Grasp}(G_{grasp}^*)$

else if $P_{push} \neq \phi$ **then**

$P_{push} \mapsto G_{push}$

$G_{push}^* = \text{argmax}_{G_{push}} Q$

$\text{Push}(G_{push}^*)$

else

$Flag \leftarrow False$

end

end

D. Manipulation Policy Module

Algorithm 1 shows the implementation of the collision-free manipulation policy. Input a set of grasp configurations G containing the angle, width, quality and category. We filter a set of grasp points P_{all} in the quality dimension $Q = \mathbb{R}^{1 \times h \times w}$ and set a to be 0.7, both to avoid missing ground-truth grasp configurations and to ensure the quality of the grasp. Then, we perform an argmax operation on the categorical dimension $C = \mathbb{R}^{3 \times h \times w}$ and index the result to obtain three sets of points $\{P_1, P_2, P_3\}$ to represent three categories. P_{grasp} is the set of grasp points without collision, which belongs to both P_{all} and P_1 . P_{push} is the set of push points that belongs to both P_{all} and P_3 . By mapping the two-dimensional point set to the four-dimensional grasp configurations, we can get G_{grasp} , G_{push} respectively. If P_{grasp} exists, we choose the grasp configuration G_{grasp}^* with the highest quality to complete the grasp action. Otherwise, the algorithm further checks for the presence of P_{push} and decides whether or not to perform the push action. After executing each action, the algorithm captures the next frame until no test tubes are visible, signaling the termination of the process.

E. Loss Function

In this study, we train our model by optimizing the minimum error between the predicted gripper configuration $G(\theta, w, q, c)$ and its corresponding label $\hat{G}(\hat{\theta}, \hat{w}, \hat{q}, \hat{c})$ at each pixel point.

1) *Grasp Quality*: The binary cross entropy function is utilized to calculate the grasp quality, which is defined as:

$$\mathcal{L}_Q = -\frac{1}{N} \sum_{n=0}^N [\hat{p}_q^n \cdot \log p_q^n + (1 - \hat{p}_q^n) \cdot \log (1 - p_q^n)] \quad (5)$$

where N is the size of output feature maps, p_q^n is the predicted quality score, $\hat{p}_q^n$ is the ground truth.

2) *Grasp Angle*: We define the objective of minimizing the binary cross entropy loss of the predicted grasp angle:

$$\mathcal{L}_\theta = -\frac{1}{N \times S} \sum_{n=0}^N \sum_{s=0}^M [\hat{p}_s^n \cdot \log p_s^n + (1 - \hat{p}_s^n) \cdot \log (1 - p_s^n)] \quad (6)$$

where $\hat{p}_s^n$ is the probability that the predicted grasp angle is within $[\frac{s}{S} \times 2\pi, \frac{s+1}{S} \times 2\pi]$ at the n position, p_s^n is the ground truth. Angle scale in our letter is set to 120.

3) *Grasp Width*: We use the binary cross entropy loss as the regression loss function, defined as (7):

$$\mathcal{L}_W = -\frac{1}{N} \sum_{n=0}^N [\hat{p}_w^n \cdot \log p_w^n + (1 - \hat{p}_w^n) \cdot \log (1 - p_w^n)] \quad (7)$$

where $\hat{p}_w^n$ is the predicted grasp width at the n position, p_w^n is the ground truth.

4) *Grasp Category*: We choose multi-class cross entropy function as loss function of category recognition:

$$\mathcal{L}_C = -\frac{1}{N} \sum_{n=0}^N \sum_{c=1}^M \hat{p}_c^n \cdot \log (p_c^n) \quad (8)$$

where M ($M = 3$) is the number of categories, p_c^n is the predicted category at the n position, $\hat{p}_c^n$ is the corresponding label. In our experiments, M is set to 3.

5) *Total Loss*: To balance the loss of each branch, the final multi-task loss is defined as follows:

$$\mathcal{L}_{total} = \lambda_1 \times \mathcal{L}_\theta + \lambda_2 \times \mathcal{L}_W + \lambda_3 \times \mathcal{L}_Q + \lambda_4 \times \mathcal{L}_C \quad (9)$$

where λ_1 , λ_2 , λ_3 and λ_4 are weight coefficients of the loss. In our study, we experimentally set $\lambda_1 = 5$, $\lambda_2 = 1$, $\lambda_3 = 1$ and $\lambda_4 = 10$.

IV. EXPERIMENT SET-UP

A. Creation of the Dataset

1) *Image Acquisition*: We create a laboratory dataset consisting of approximately 1000 RGB images containing both simple and complex arrangements. These images are captured using an Intel RealSense D435i camera and have a size of 640×480 pixels. In our training, we randomly select 90% as the training set and the remaining 10% as the test set. In order to improve the actual generalization performance of the model, we simulate as much as possible the various scenarios encountered during the deployment on the training set. In addition, we use a technique called color jittering augmentation to increase the generality of the model, which includes adjustments to saturation, hue shift, and contrast.

2) *Data Annotation*: Traditional object detection annotation tools are not suitable for pixel-level annotation. While commonly used pixel-level segmentation annotation tools can provide pixel-level annotation, the entire process can be time-consuming. Thus, Grasplabel, a Python program, is developed to efficiently annotate the grasping attributes at the pixel level. The program allows manual labelling of the pixel sets of the graspable regions and the corresponding category, and calculation of the corresponding grasp angle and width.



Fig. 4. Grasp detection results on the test tube dataset. The first row visualizes the collected images of different numbers of test tubes. The second row visualizes the grasp points of different categories. The third row visualizes the grasp configuration with the highest grasp quality scores.

B. Set up and Training Details

The experiments are conducted on the 6-DOF Xarm6 robot with a paired two-fingered parallel gripper by UFACTORY. We use the eye-to-hand calibration system with the Intel RealSense Depth Camera D435i mounted on a fixed stand to obtain scene images.

The execution times for our proposed network are measured on a system running Ubuntu 20.04 with an Intel Core i9-12900KF and an NVIDIA GeForce GTX 3090 graphics card with CUDA 11.7. It uses the Adam optimizer with a learning rate of 0.001. The neural network is trained end-to-end for 300 epochs, with learning rate decaying by a factor of 0.5 at epochs 100, 200, and 300.

C. Evaluation Metric for Grasp Detection

According to the rectangle metric proposed by Jiang et al. [22], a grasp is judged to be correct if it satisfies the following two conditions:

1) *Angle Index*: The difference in angle between the predicted grasp angle and the label angle is less than 30° .

2) *The Jaccard Index*: The intersection over union (IoU) of the predicted grasp (G_p) and the label grasp (G_l) is greater than 0.25, as shown in (10).

$$J(G_p, G_l) = \frac{|G_p \cap G_l|}{|G_p \cup G_l|} > 0.25 \quad (10)$$

V. EXPERIMENTS

A. Grasp Detection

Fig. 4 shows some qualitative grasp detection results for different numbers of test tubes. We divide the predicted grasp regions into three categories using the grasp category output head, which provides crucial semantic information to guide

TABLE I
ABLATION STUDIES FOR GRASP DETECTION

Baseline	REASP-R	REASP-P	Detection accuracy(%)
✓			92.47
✓	✓		94.62
✓		✓	95.70
✓	✓	✓	96.77

TABLE II
ROBUSTNESS EVALUATION OF GRASP POINT PREDICTION ON TUBE DATASET
UNDER DIFFERENT CONDITIONS

Tube type	Plastic tray		Black desktop		Slightly rugged		Steel tray	
	Bright	Dark	Bright	Dark	Bright	Dark	Bright	Dark
I	0.99	0.98	0.98	0.96	0.95	0.90	0.92	0.88
II	0.97	0.96	0.95	0.94	0.93	0.89	0.88	0.86
III	0.97	0.95	0.95	0.94	0.92	0.89	0.88	0.87

collision-free grasping and pushing actions. In these scenarios, the predicted set of grasping points almost overlaps with the ground truth values. All the grasp detection results show that the proposed network can accurately generate a set of grasp configurations and achieve excellent performance in cluttered environments.

We further train models with different network settings on the tube dataset to explore the impacts other modules exert on the algorithms. The ablation studies are divided into REASP-Parallel, REASP-Residual. For the baseline of our network, we use ResNet-101 and Atrous Spatial Pyramid. The four additional parallel branches with group convolution and residual connection are removed directly, but the fusion module is retained. The evaluation metric used is the accuracy on the test set, corresponding to a Jacquard index of 0.25 and an angle index of 30°. Table I summarizes the experimental results, which shows the optimized performance of improvement modules for grasp detection. The baseline gets accuracy of 96.77%. Compared to the baseline, the accuracy is enhanced by 2.15% and 3.23% via REASP-P and REASP-R, respectively. Combining the two improvements, the accuracy is increased by 4.3%. There are compensations for more parallel branches of REASP with different dilation rates, resulting in the perception of a larger context.

B. Robustness and Generalization Evaluation

In order to validate the robustness of our method for real medical scenarios, we recapture image samples under different conditions and conduct a robustness experiment. The evaluations are conducted under three specific conditions: illumination, background, and test tube type. The illumination conditions consist of bright and dark illumination settings. The background includes plastic tray, stainless steel tray, black desktop and slightly rugged desktop. The different tube types differ mainly in the color of the caps. The combination of the conditions creates 24 imaging scenarios, with each scenario tested 100 times. In each test, we assess whether the output grasp point is a collision-free grasp point. The results of grasp point prediction are presented in Table II, and the visualization of optimal collision-free grasps under different environments is shown in

TABLE III
EVALUATION RESULTS OF DIFFERENT METHODS ON THE CORNELL DATASET

Authors	Method	Accuracy(%)	Time(ms)	Input data format
Kumra <i>et al.</i> [30]	Resnet-50	89.2	103	RGB-D
Asif <i>et al.</i> [21]	GraspNet	90.6	24	RGB-D
Song <i>et al.</i> [31]	GRPN	95.6	-	RGB
Wang <i>et al.</i> [32]	GPWRG	94.4	8	RGB-D
Chu <i>et al.</i> [20]	FasterR-CNN	96.0	120	RGB-D
Kumra <i>et al.</i> [6]	GR-ConvNet	97.7	20	RGB
Cao <i>et al.</i> [33]	Efficient Grasp	97.8	6	RGB
Yu <i>et al.</i> [34]	SE-ResUNet	98.2	25	RGB-D
Ours	REASP-Grasp	98.7	20	RGB

TABLE IV
RESULTS OF REAL-WORLD GRASPING EXPERIMENT

Arrangement cases	Complexity index	Completion	SR	Action efficiency
random	I	100	100(90/90)	98.9
	II	100	99.3(150/151)	97.4
	III	100	99.1(210/212)	96.3
	IV	100	99.0(270/273)	96.4
	V	100	98.5(330/335)	95.9
challenging	III	96.7	95.4(206/216)	82.0
	IV	96.7	95.4(267/280)	82.8
	V	93.3	94.6(318/336)	81.3

* Complexity index is related to the number of tubes per unit space and how closely the tubes are arranged.

Fig. 5. The data demonstrate that variations in illumination and test tube color have minimal impact on the predicted performance across all scenarios. Since the RGB-based method itself can overcome slightly curved surfaces, our method performs well on slightly rugged desktop. Notably, there is a significantly performance degradation when the background changes to a steel tray. However, this degradation typically remains within an acceptable range. We speculate that the reason for the performance degradation shown could be due to the reflective nature of the steel tray, which appears as high highlights in the RGB images and thus affecting the prediction of the grasping results.

The network part of our proposed method can be trained separately for the usual planar grasping. To verify the generalization of the network to other objects, we compare our model with representative methods on the public Cornell dataset. Table III shows that our model achieves a high detection accuracy of 98.7%, which is better than other state-of-the-art algorithms. The inferred speed is 20 ms per image, which guarantees its real-time detection applications. Most of the algorithms are not open source, so we do not compare them on the test tube dataset.

C. Robotic Grasping

In this section, we conduct real-world grasping experiment on our robot platform to validate the performance in different scenarios, including random scenes and challenging scenes. In random scenes, objects are randomly positioned, and the focus is on maximizing grasping actions while minimizing the need for pushing actions. Challenging scenes involve manual placement of objects in close proximity, requiring at least one pushing



Fig. 5. Obtained images under different conditions for robustness evaluation. The red arrow visualizes the predicted best collision-free grasp point.

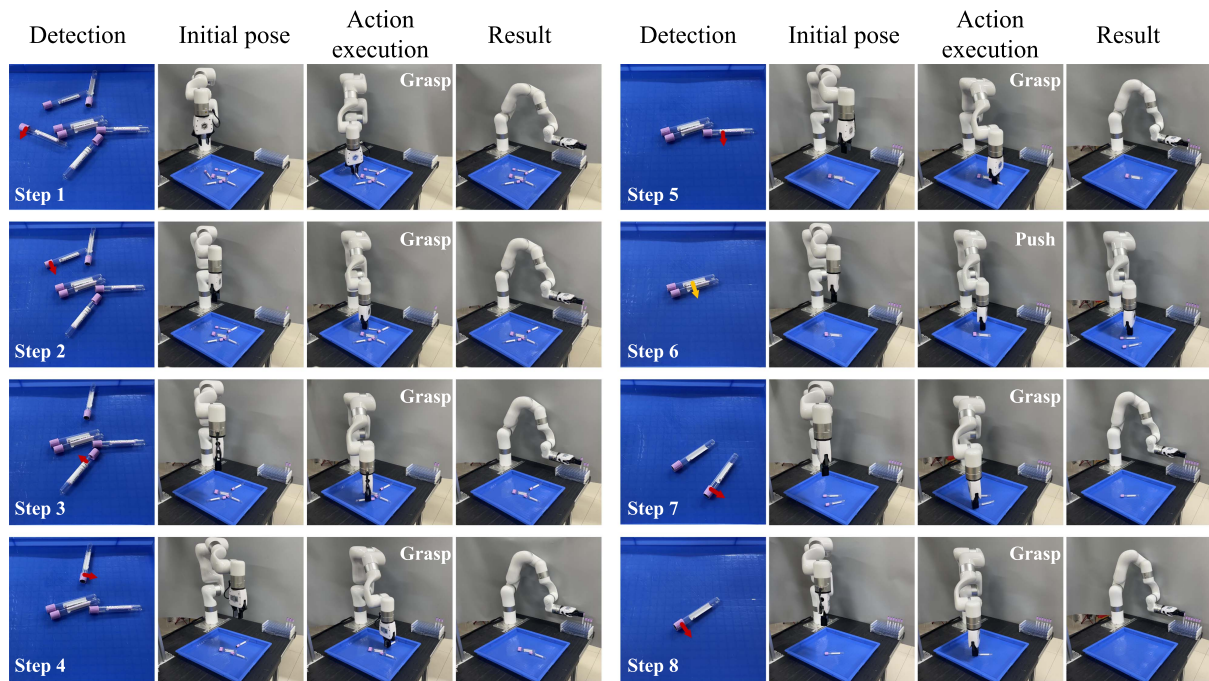


Fig. 6. Robotic grasping experiments on the Xarm6 robot operating platform for challenging scenarios (corresponding complexity index is III). The visualized grasping results in the figure correspond to the estimated optimal collision-free grasp (red arrow indicates grasping action, yellow arrow indicates pushing action), the initial pose of the robot, the action execution and the successful picking, respectively.

action in each trial, as shown in Fig. 6. Evaluation metrics are measured based on widely used criteria, as follows:

1) *Completion*: The percentage of completion rate after performing N test runs. If the robot successfully grasps all objects and failures less than three times, the task is considered completed.

2) *Grasp Success Rate (SR)*: Ratio of successful grasps to total. The parallel jaw gripper reaches a predetermined position to grip and lift the object, i.e. a successful grasp.

3) *Action Efficiency*: Action efficiency is defined as the ratio of the number of objects to be grasped in the scene to the total number of actions, which is a metric used to measure the efficiency of the method to complete a grasping task.

To validate the performance of the proposed method for collision-free grasp detection in cluttered scenes, we conduct

practical experiment with different levels of complexity, which consists of 7 trials with 1893 grasp actions and 151 pushing actions. In each trial, we perform 30 grasp tasks under the same conditions while ensuring different initial positions.

Table IV reports the evaluation of our approach in different scenes in terms of the real-world metrics. It can be observed that in random scenes, even with an increase in the number of test tubes, our approach is capable of handling the grasping task with a grasp success rate of over 98% and an action efficiency of over 95%. In challenging scenes, our approach maintains high performance across various evaluation metrics. The slightly lower action efficiency of around 80% can be attributed to the increased number of pushing actions required in complex scenarios, which is related to the definition itself. In practical experiments, the pushing action demonstrates good performance,

which effectively separates the test tubes and creates space of collision-free grasping. The results indicate that our method performs well in cluttered environments.

VI. CONCLUSION

In this letter, we introduce an improved dense pixel-level prediction architecture for medical device grasp detection, and construct a practical robotic grasping platform for evaluation. To avoid the confusion of the grasp quality score prediction, we propose the AGFC-model to additionally assign a grasp category label to each pixel. Then, we propose an end-to-end pixel-level segmentation network to generate the AGFC-model and perform modular improvements to increase the accuracy and efficiency of grasp detection. Finally, we design a manipulation policy to guide the real robot in collision-free grasping. Through the experiments on the established tube dataset and Cornell dataset, we verify that the proposed network can achieve performance for grasp detection comparable to that of state-of-the-art methods. Furthermore, the experiments confirm that our method has a good generalization performance under different conditions and performs well in real-world grasping and placement tasks.

REFERENCES

- [1] Z. Wang, "Vision-based robot arm grasping in medical region," *Highlights Sci., Eng. Technol.*, vol. 43, pp. 261–264, 2023.
- [2] A. Yoganandhan, G. R. Kanna, S. Subhash, and J. H. Jothi, "Retrospective and prospective application of robots and artificial intelligence in global pandemic and epidemic diseases," *Vacunas (English Edition)*, vol. 22, no. 2, pp. 98–105, 2021.
- [3] J. Bohg, A. Morales, T. Asfour, and D. Kragic, "Data-driven grasp synthesis—A survey," *IEEE Trans. Robot.*, vol. 30, no. 2, pp. 289–309, Apr. 2014.
- [4] L. Berscheid, P. Meißner, and T. Kröger, "Robot learning of shifting objects for grasping in cluttered environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 612–618.
- [5] M. Laskey et al., "Robot grasping in clutter: Using a hierarchy of supervisors for learning from demonstrations," in *Proc. IEEE Int. Conf. Automat. Sci. Eng.*, 2016, pp. 827–834.
- [6] S. Kumra, S. Joshi, and F. Sahin, "Antipodal robotic grasping using generative residual convolutional neural network," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 9626–9633.
- [7] V. Satish, J. Mahler, and K. Goldberg, "On-policy dataset synthesis for learning robot grasping policies using fully convolutional deep networks," *IEEE Robot. Automat. Lett.*, vol. 4, no. 2, pp. 1357–1364, Apr. 2019.
- [8] A. Depierre, E. Dellandréa, and L. Chen, "Jacquard: A large scale dataset for robotic grasp detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2018, pp. 3511–3516.
- [9] J. Mahler et al., "Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics," *Robot.: Sci. Syst.*, 2017.
- [10] S. Wang, Z. Zhou, and Z. Kan, "When transformer meets robotic grasping: Exploits context for efficient grasp detection," *IEEE Robot. Automat. Lett.*, vol. 7, no. 3, pp. 8170–8177, Jul. 2022.
- [11] H. Zhang, Y. Wu, E. Demeester, and K. Kellens, "Big-Net: Deep learning for grasping with a bio-inspired soft gripper," *IEEE Robot. Automat. Lett.*, vol. 8, no. 2, pp. 584–591, Feb. 2023.
- [12] J. Lundell, F. Verdoja, and V. Kyrki, "DDGC: Generative deep dexterous grasping in clutter," *IEEE Robot. Automat. Lett.*, vol. 6, no. 4, pp. 6899–6906, Oct. 2021.
- [13] M. Sundermeyer, A. Mousavian, R. Triebel, and D. Fox, "Contact-GraspNet: Efficient 6-DoF grasp generation in cluttered scenes," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 13438–13444.
- [14] J. Lundell, F. Verdoja, and V. Kyrki, "Beyond top-grasps through scene completion," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 545–551.
- [15] D. Morrison, P. Corke, and J. Leitner, "Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach," *Robot.: Sci. Syst.*, MIT Press-Journals, 2018.
- [16] J. Mahler and K. Goldberg, "Learning deep policies for robot bin picking by simulating robust grasping sequences," in *Proc. Conf. Robot Learn.*, 2017, pp. 515–524.
- [17] J. Mahler et al., "Learning ambidextrous robot grasping policies," *Sci. Robot.*, vol. 4, no. 26, 2019, Art. no. eaau4984.
- [18] K. Rahardja and A. Kosaka, "Vision-based bin-picking: Recognition and localization of multiple complex objects using simple visual cues," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 1996, vol. 3, pp. 1448–1457.
- [19] X. Yan et al., "Data-efficient learning for sim-to-real robotic grasping using deep point cloud prediction networks," 2019, *arXiv:1906.08989*.
- [20] F.-J. Chu, R. Xu, and P. A. Vela, "Real-world multiobject, multigrasp detection," *IEEE Robot. Automat. Lett.*, vol. 3, no. 4, pp. 3355–3362, Oct. 2018.
- [21] U. Asif, J. Tang, and S. Harrer, "GraspNet: An efficient convolutional neural network for real-time grasp detection for low-powered devices," in *Proc. Int. Joint Conf. Artif. Intell.*, 2018, vol. 7, pp. 4875–4882.
- [22] Y. Jiang, S. Moseson, and A. Saxena, "Efficient grasping from RGBD images: Learning using a new rectangle representation," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2011, pp. 3304–3311.
- [23] I. Lenz, H. Lee, and A. Saxena, "Deep learning for detecting robotic grasps," *Int. J. Robot. Res.*, vol. 34, no. 4/5, pp. 705–724, 2015.
- [24] L. Pinto and A. Gupta, "Supersizing self-supervision: Learning to grasp from 50 k tries and 700 robot hours," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2016, pp. 3406–3413.
- [25] D. Wang, C. Liu, F. Chang, N. Li, and G. Li, "High-performance pixel-level grasp detection based on adaptive grasping and grasp-aware network," *IEEE Trans. Ind. Electron.*, vol. 69, no. 11, pp. 11611–11621, Nov. 2021.
- [26] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [27] R. Araki, T. Onishi, T. Hirakawa, T. Yamashita, and H. Fujiyoshi, "MT-DSSD: Deconvolutional single shot detector using multi task learning for object detection, segmentation, and grasping detection," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 10487–10493.
- [28] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 801–818.
- [29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [30] S. Kumra and C. Kanan, "Robotic grasp detection using deep convolutional neural networks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 769–776.
- [31] Y. Song, L. Gao, X. Li, and W. Shen, "A novel robotic grasp detection method based on region proposal networks," *Robot. Comput.- Integr. Manuf.*, vol. 65, 2020, Art. no. 101963.
- [32] S. Wang, X. Jiang, J. Zhao, X. Wang, W. Zhou, and Y. Liu, "Efficient fully convolution neural network for generating pixel wise robotic grasps with high resolution images," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2019, pp. 474–480.
- [33] H. Cao, G. Chen, Z. Li, Q. Feng, J. Lin, and A. Knoll, "Efficient grasp detection network with Gaussian-based grasp representation for robotic manipulation," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 3, pp. 1384–1394, Jun. 2023.
- [34] S. Yu, D.-H. Zhai, Y. Xia, H. Wu, and J. Liao, "SE-ResUNet: A novel robotic grasp detection method," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 5238–5245, Apr. 2022.